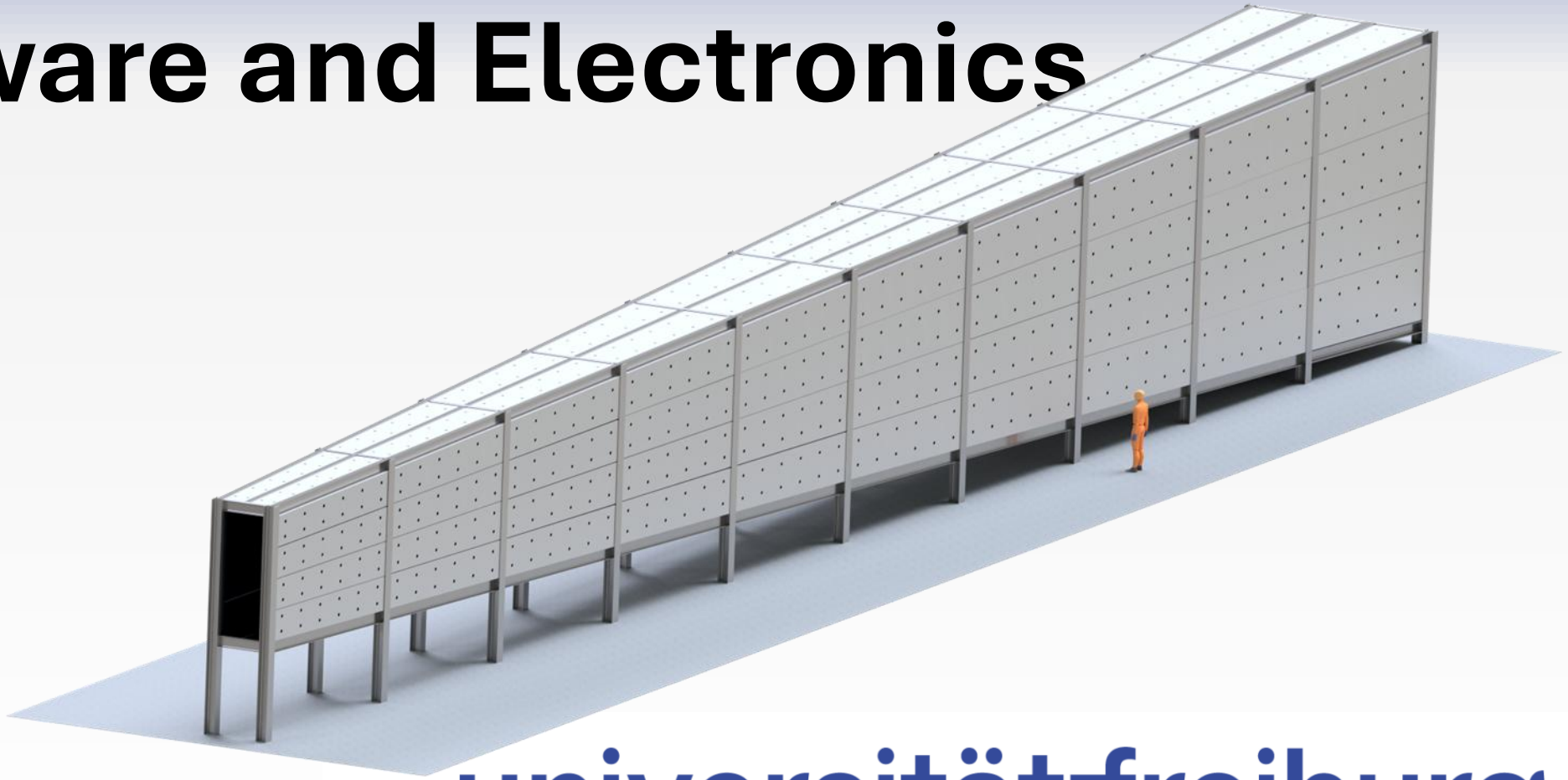
Large liquid scintillator detectors in SHiP
Hardware and Electronics

SHiP
Search for Hidden Particles



Tilman Rock
tilman.rock@uranus.uni-freiburg.de

universität freiburg

Detection mechanism of SHiP-SBT

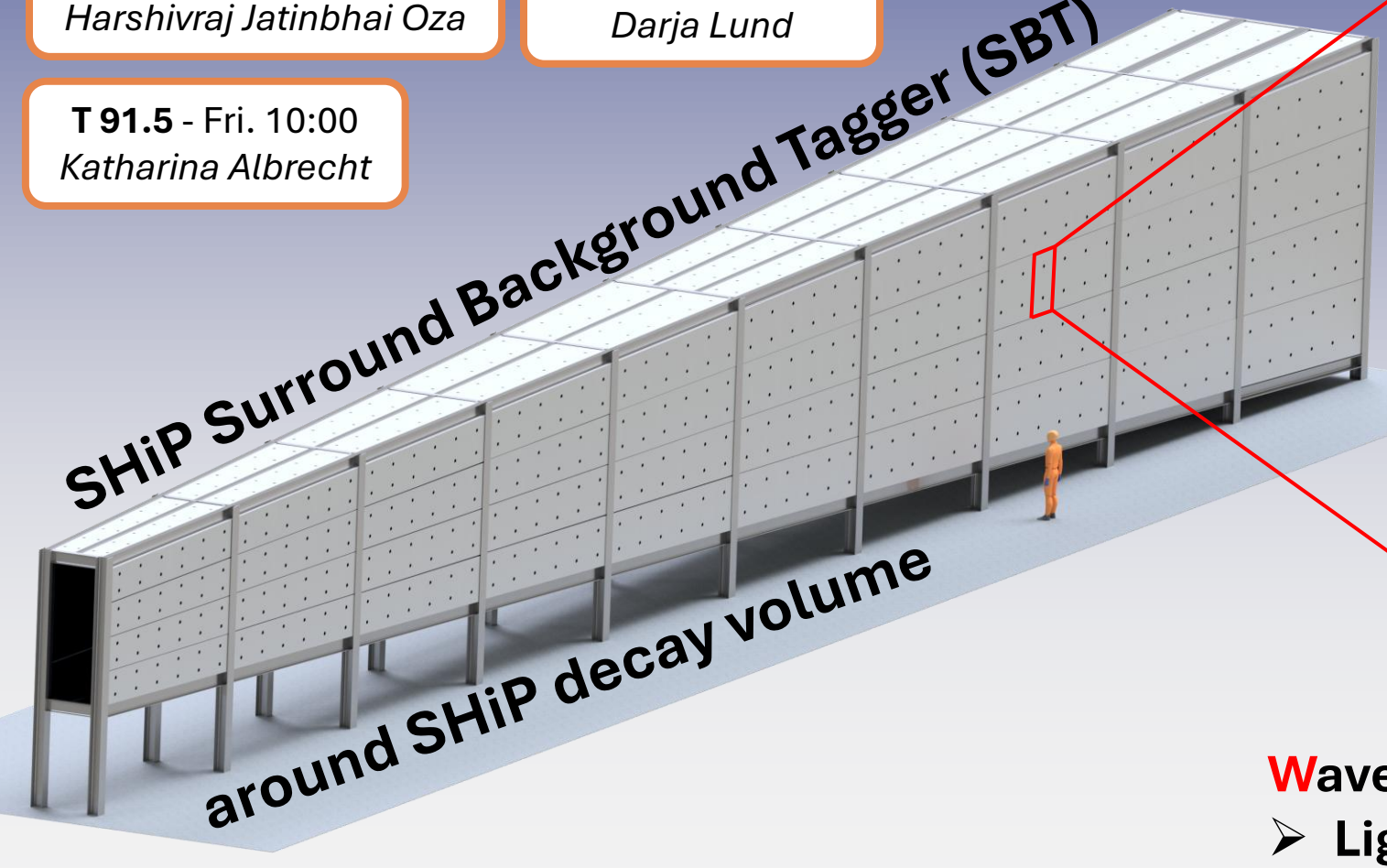
T 48.2 - Wed. 16:30
Jasmin Weiß

T 53.9 - Wed. 18:15
Hannes Braune

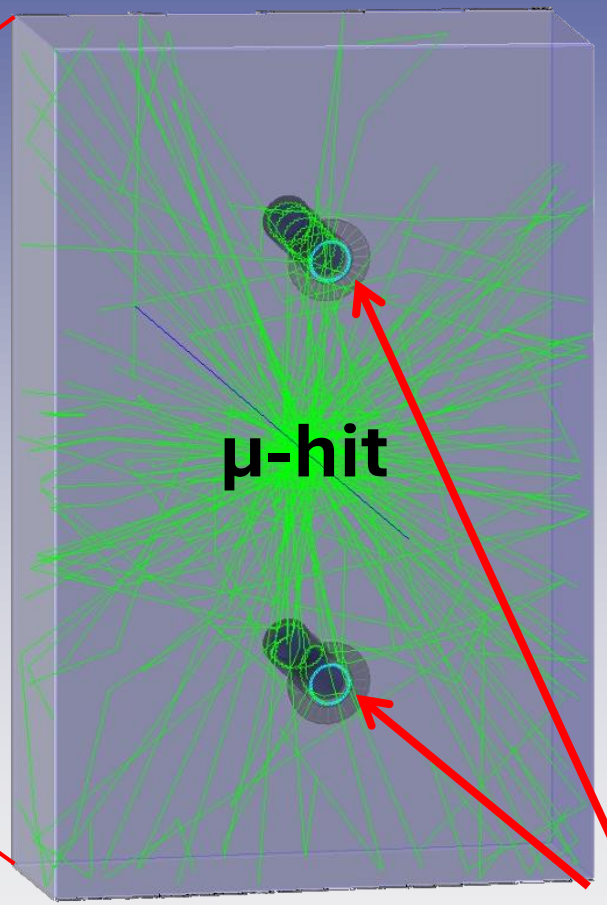
T 69.4 - Thu. 17:00
Harshivraj Jatinbhai Oza

T 91.4 - Fri. 09:45
Darja Lund

T 91.5 - Fri. 10:00
Katharina Albrecht



Liquid scintillator filled cells



WOM:

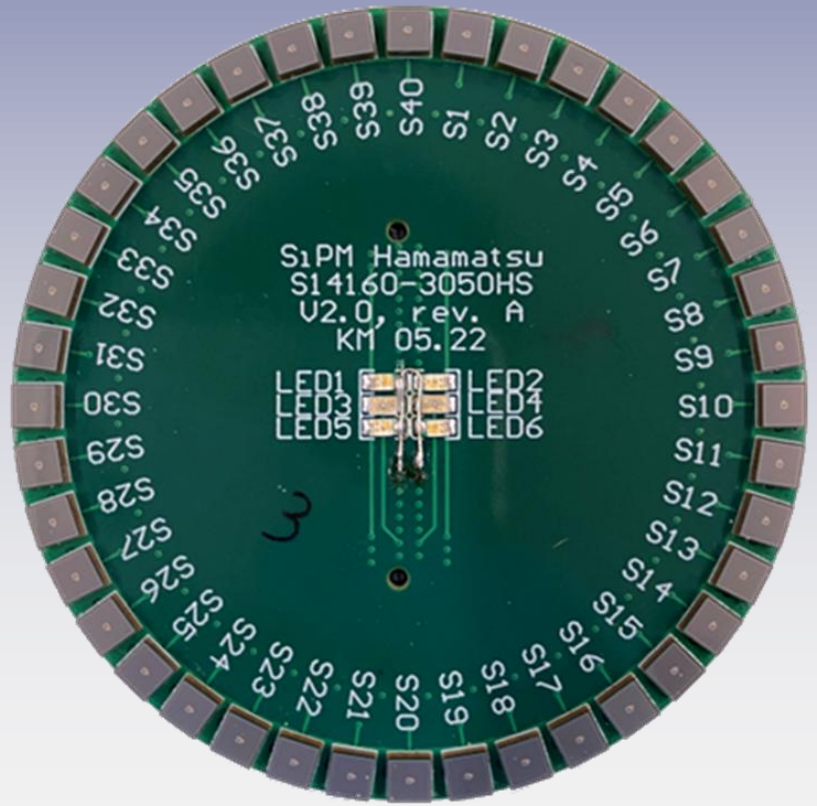
Wavelength shifting **O**ptical **M**odules (tubes)

- Light collection and λ-shifting

WOM Readout

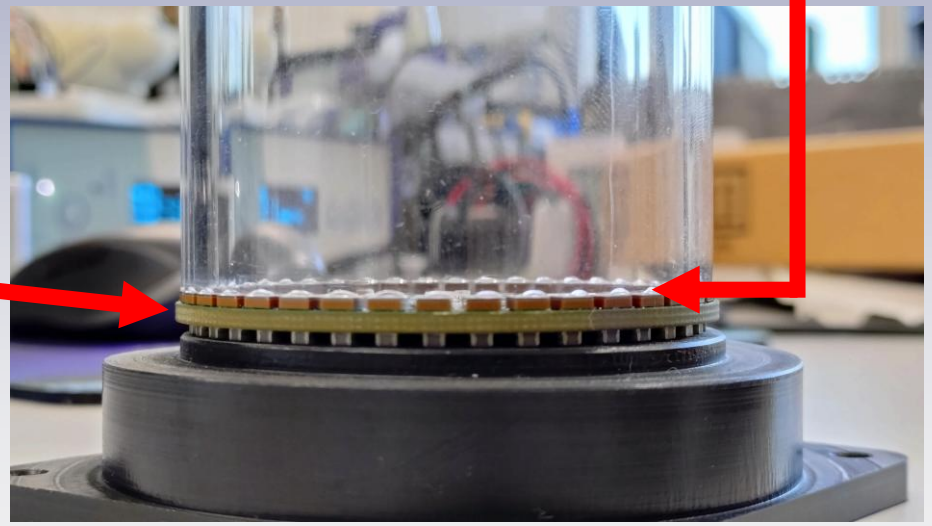
T 53.8 - Wed. 18:00
Ida Wöstheinrich

Round PCB with **40 Silicon Photomultipliers** in **8 Channels** to detect incoming light (SiPM-Board)



SiPMs: Hamamatsu S14160-3050HS

Optical grade silicone grease or epoxy resin in between



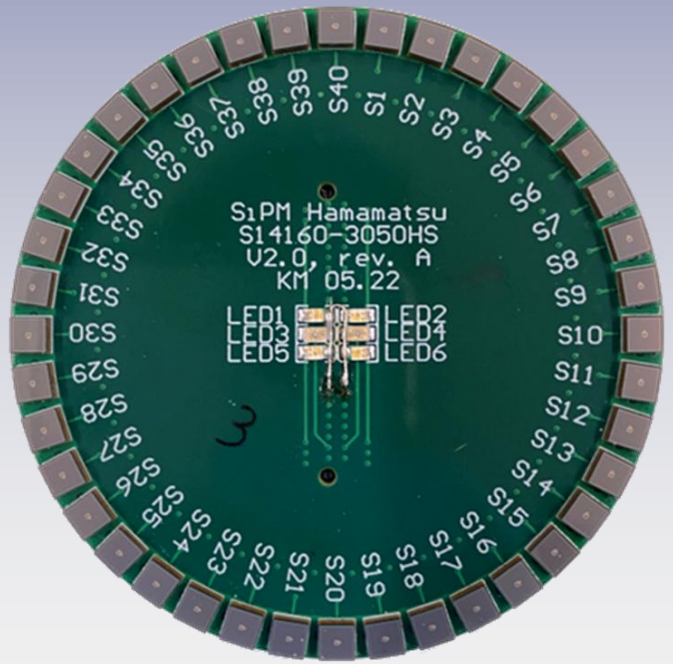
WOM optically coupled to SiPMs

Grease: Korasilon M-S 2-270 **Epoxy: EPO-TEK 301-2**

Two important technical details

Stable operation with changing environments

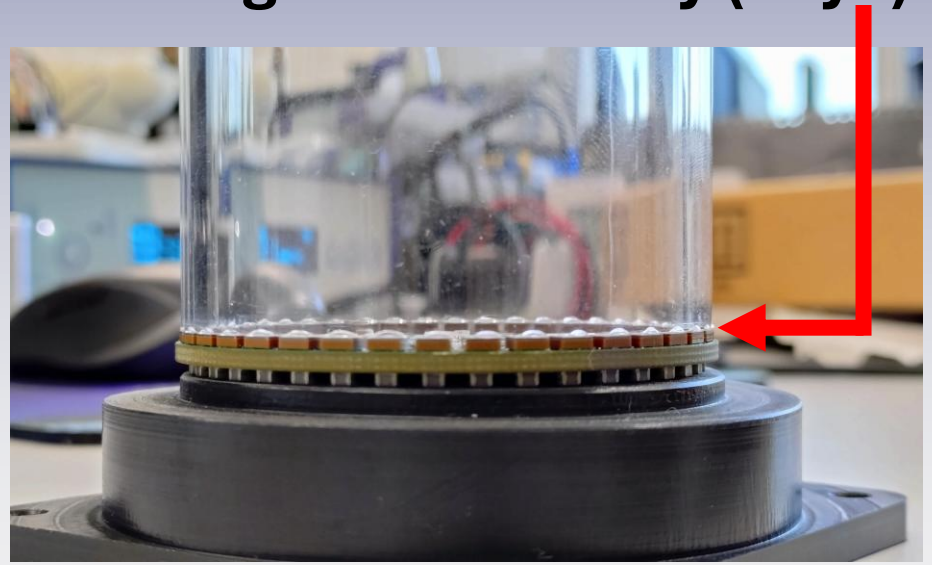
- Same light signal should produce same electrical signal



Temperature compensation

Want highest detector efficiency

- Get as much light from the WOM tube into SiPMs
- Long term stability (15y+)

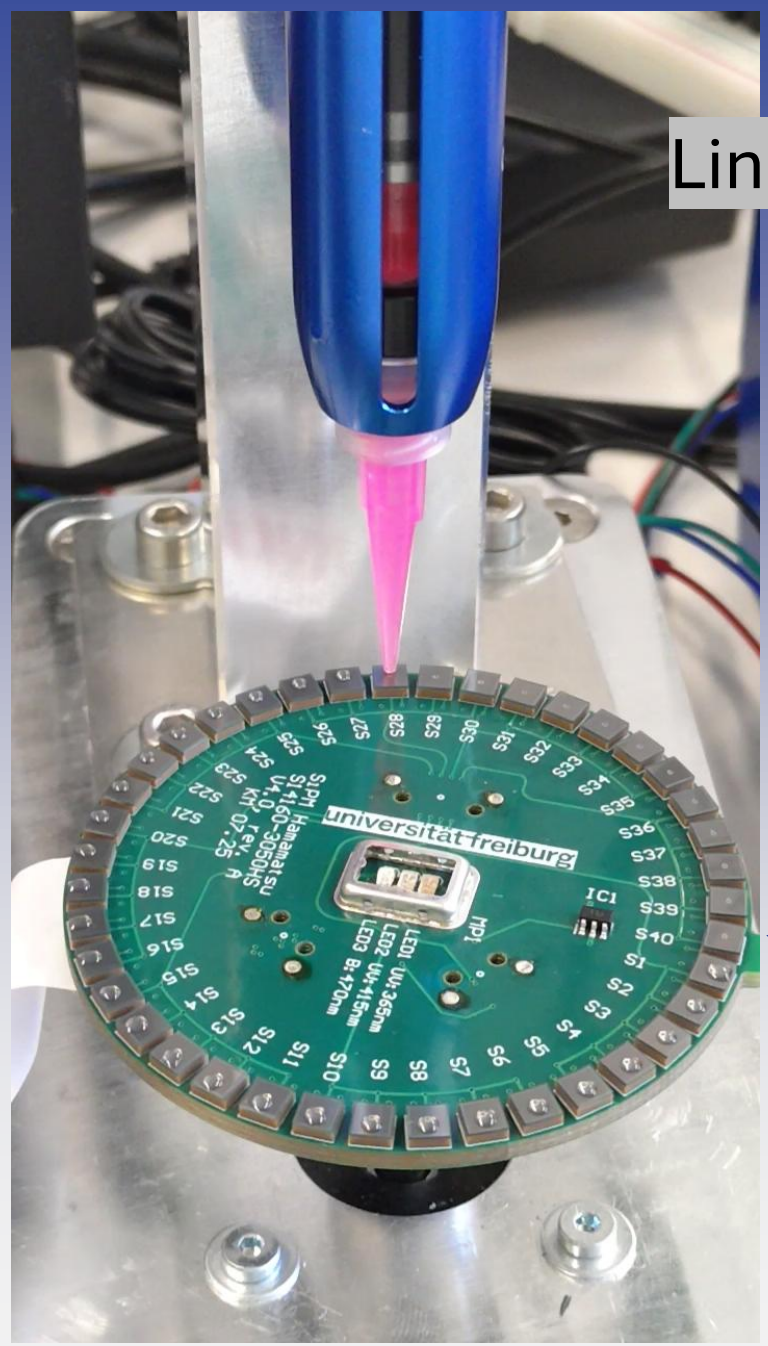


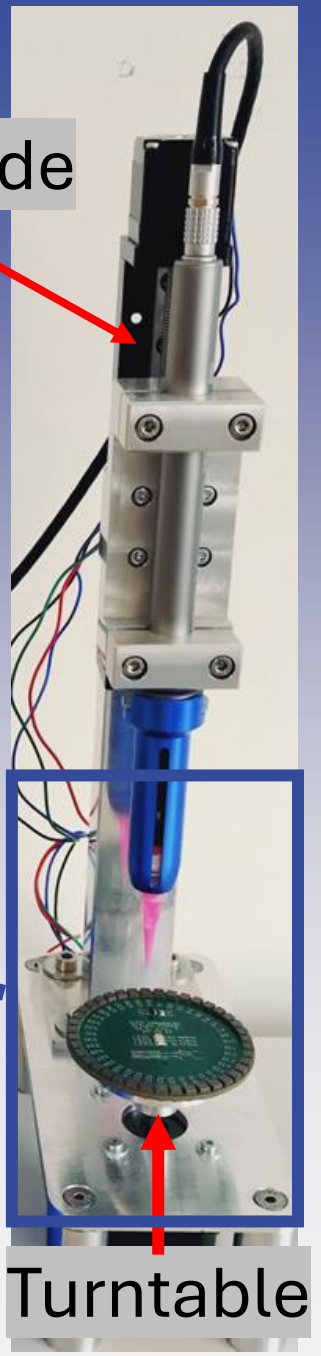
Controlled epoxy resin application

Epoxy application machine

Syringe: PRECIFLUID PDS-DRU3CC

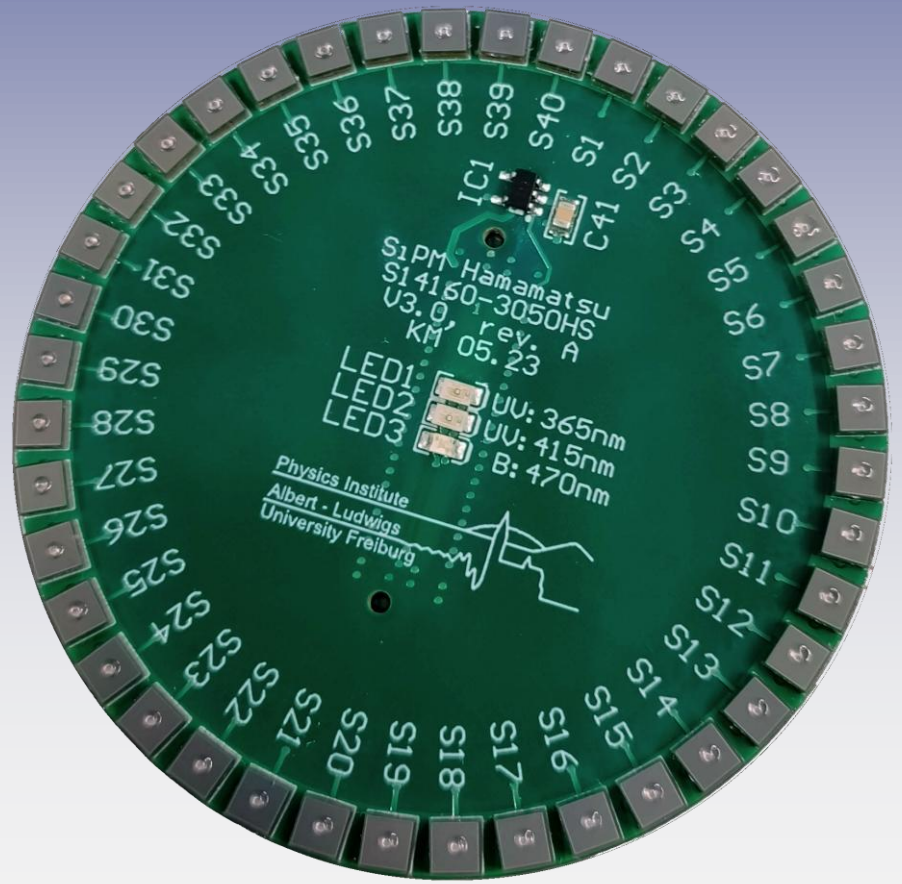
- Automatic syringe filled with optical epoxy (vacuum-debubbled)
- Raspberry Pico Program:
  - Lower syringe
  - Dispense epoxy
  - Raise syringe
  - Rotateboard to next SiPM
- 750 µg epoxy per chip in a total of ~2 min



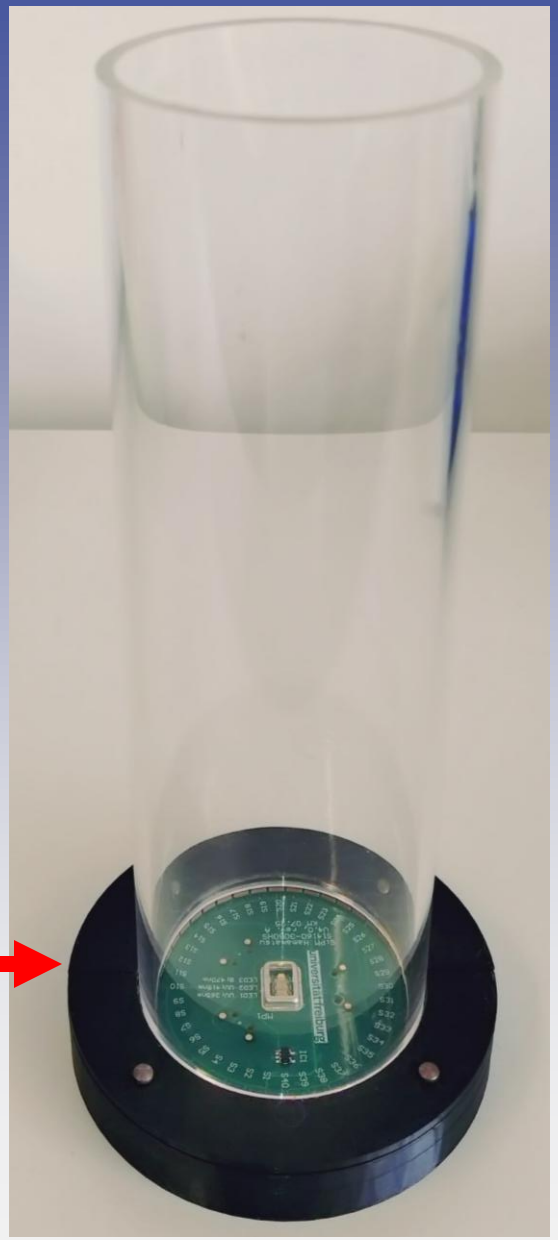


Centering of WOM tube

SiPMs with applied epoxy resin

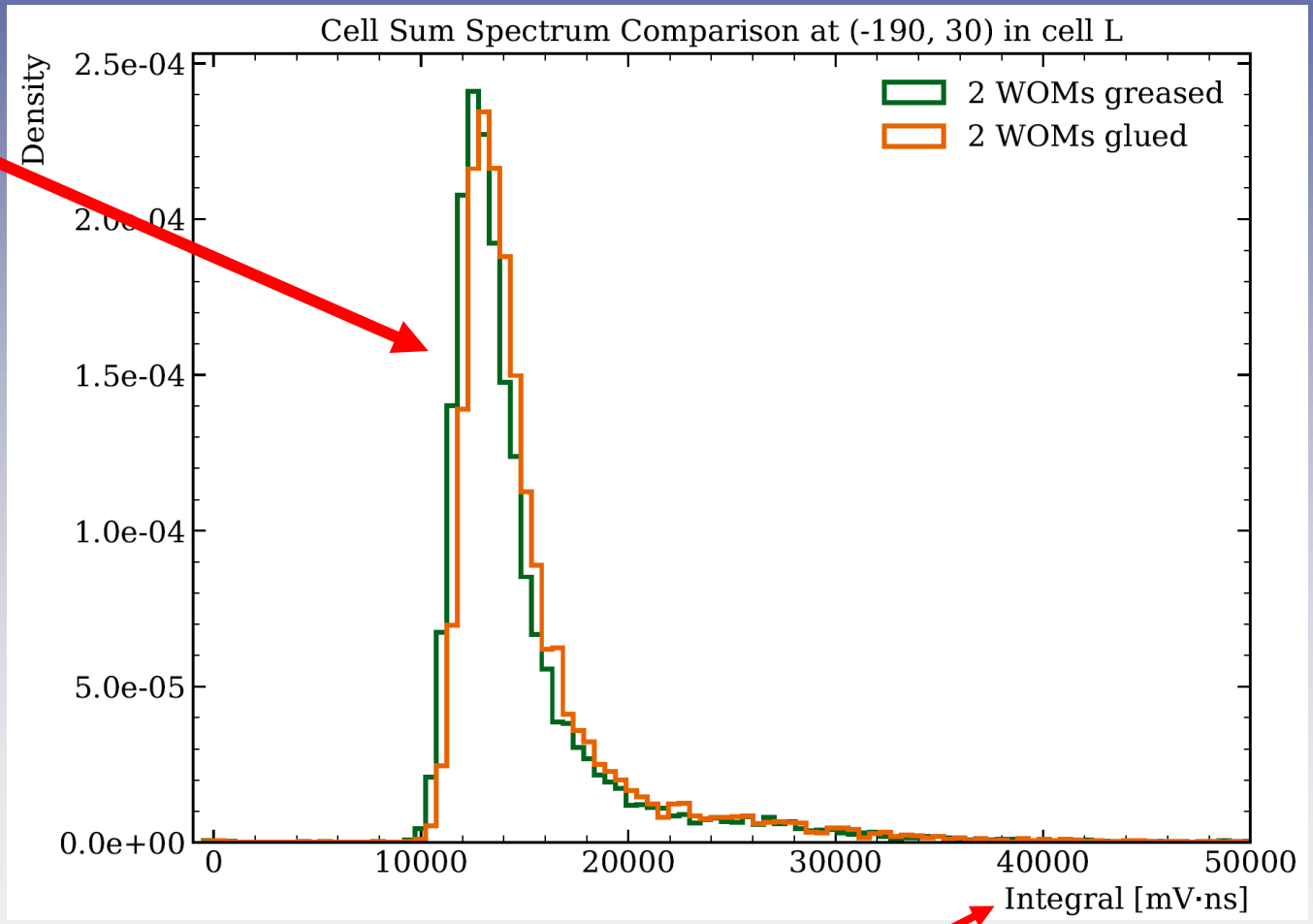


Put into concentric alignment tool



Choosing epoxy resin over grease

- **Similar coupling performance!**
- **Higher longterm stability, no creeping of the fluid along the WOM tube over time**
- **Repeatability: SiPM-WOM combination can be moved without disturbing the coupling!**

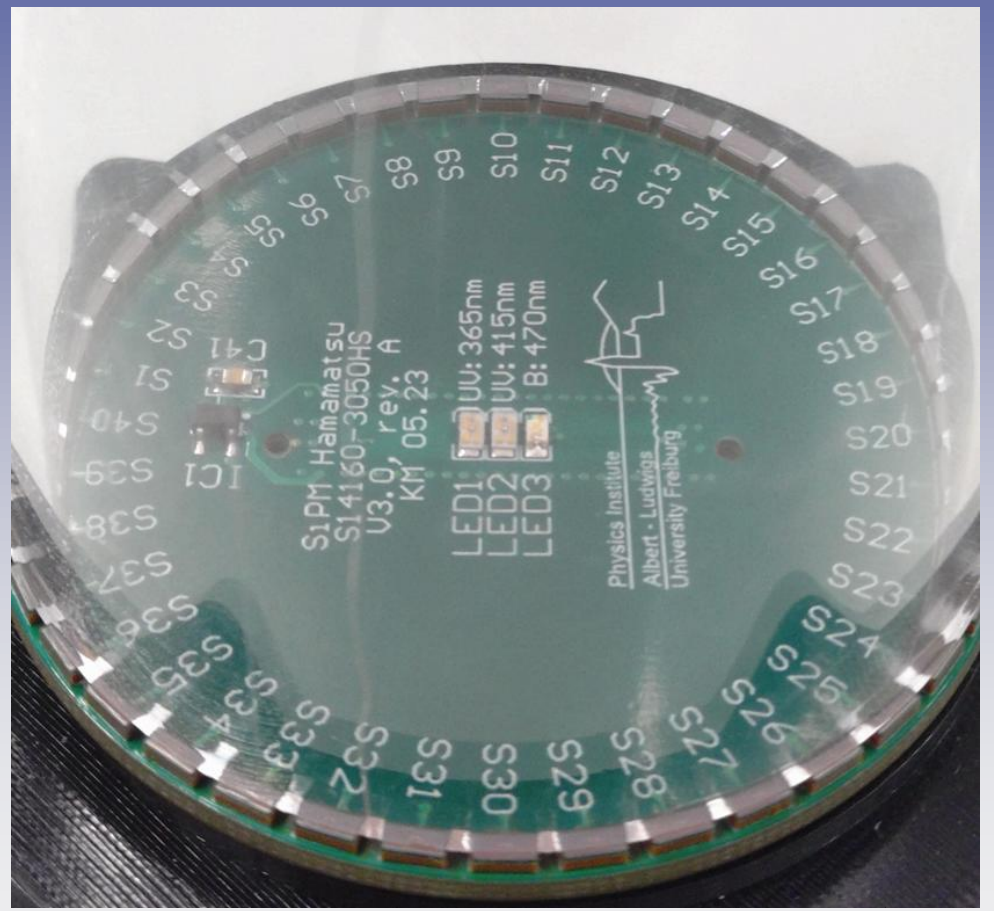


Paul Luther

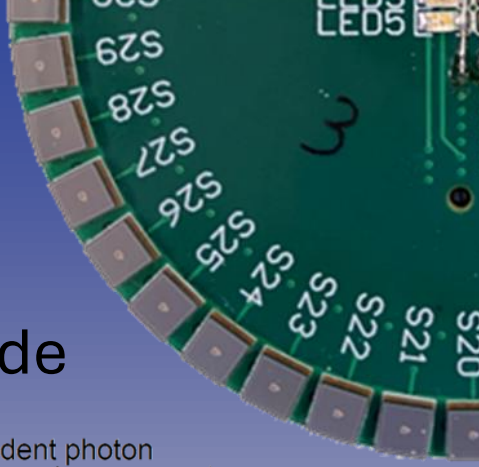
Integrated SiPM signals

Nice! We have our optical coupling.

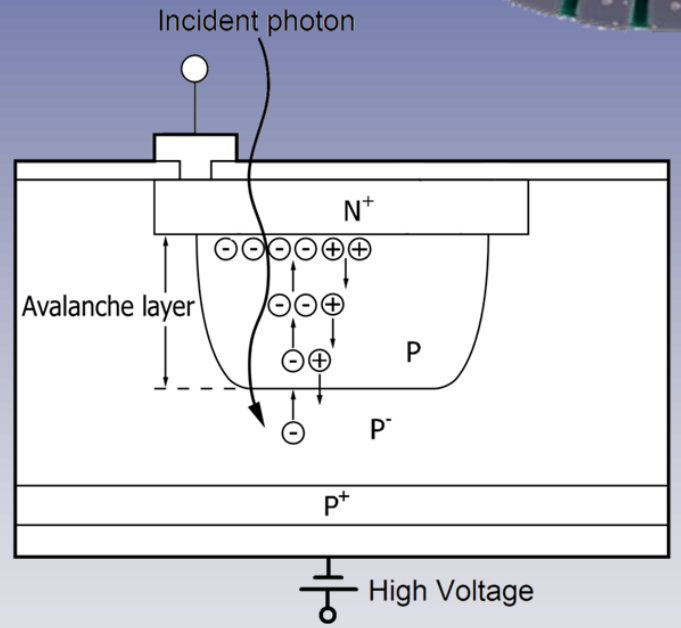
How to optimize our readout to work in different temperatures now?



SiPM-Breakdown voltage



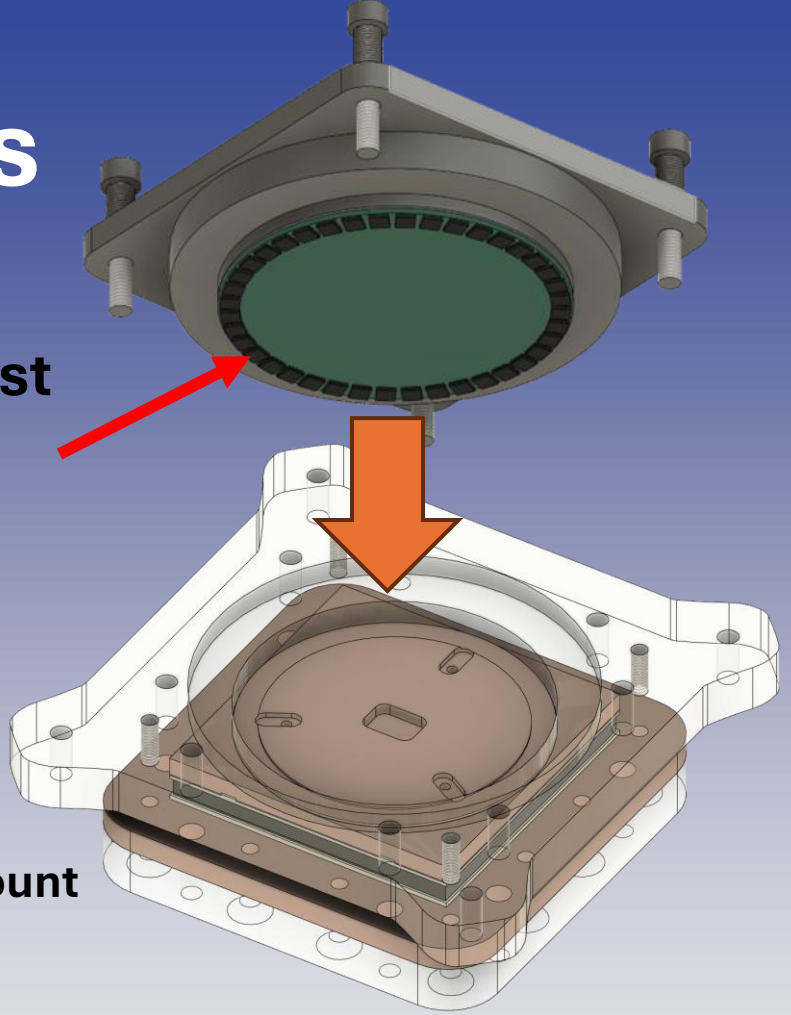
- **Breakdown voltage V_b** → Minimum reverse voltage so **avalanches** inside single photon avalanche diodes (SPADs) can occur



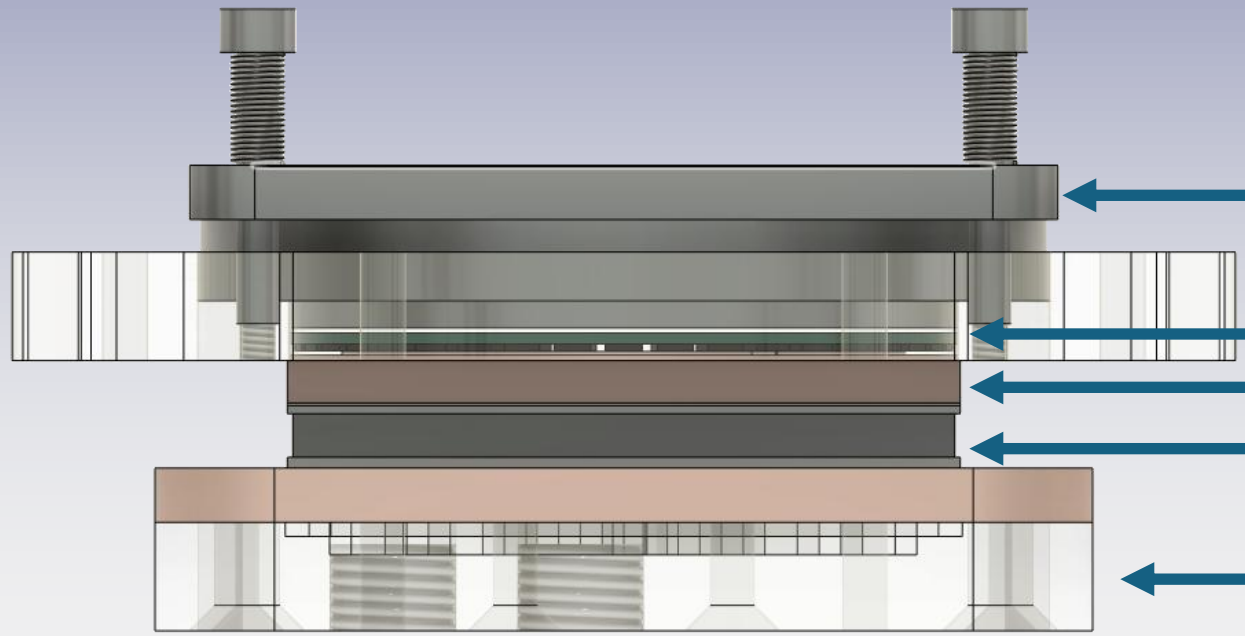
- **Overvoltage V_{ov}:** Excess reverse bias over V_b : $\mathbf{V_{ov} = V_{bias} - V_b}$
  - Intrinsic **Gain ∝ V_{ov}**
- **V_b depends on temperature**
  - Higher temperature
    → smaller mean free path of avalanche electrons → less energy gain before impact
- **Gain ∝ V_{ov} and V_{ov} ∝ T → Gain ∝ T**
  - Solution: Stabilize Gain by **adjusting V_b according to Temperature!**

How to measure this?

Temperature control mechanics



SiPM lightly pressed against copper plate with thermal pad in between



Pressure Mount

SiPM-Board

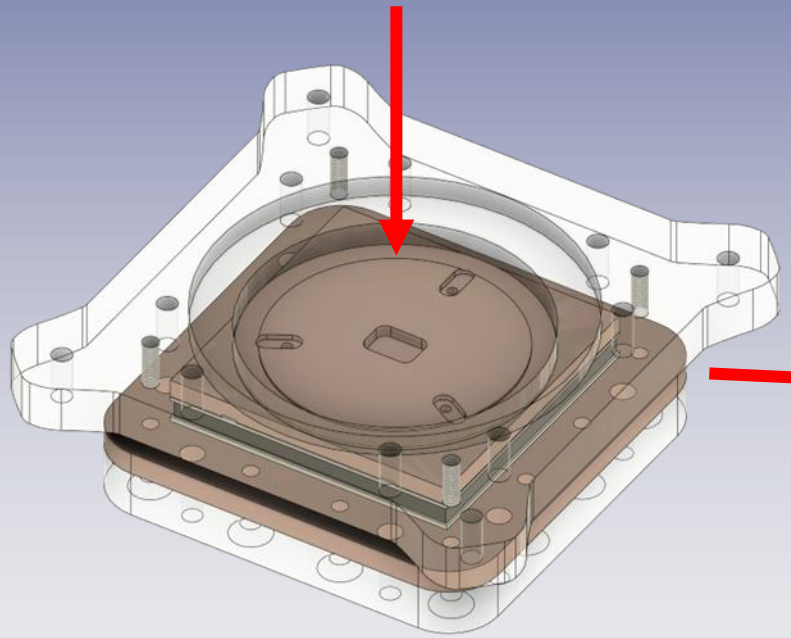
Copper heatsink with 3 PT-1000 temp. sensors

Peltier-Element

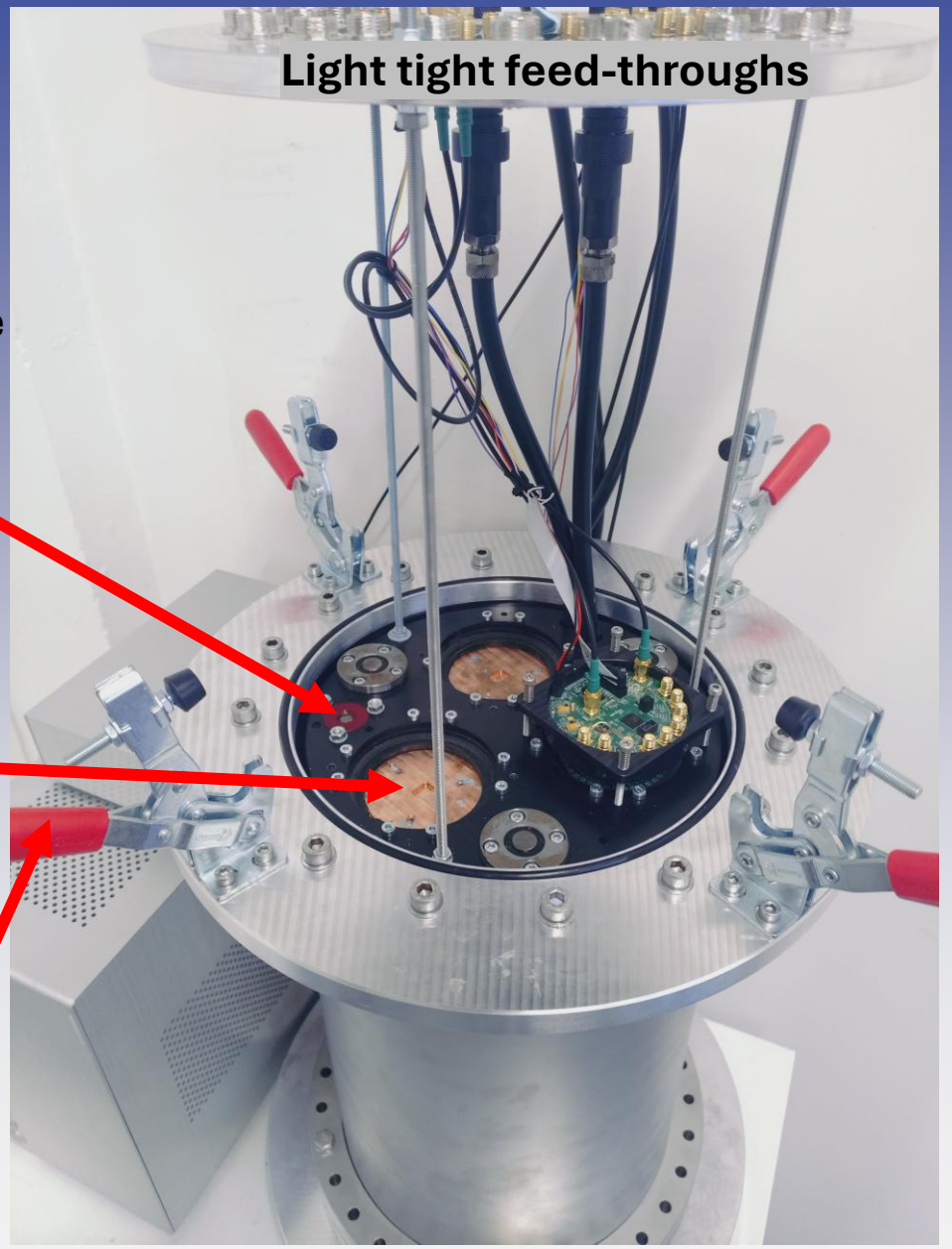
Watercooler for heat exchange

Dark vessel

3 individual SiPM cooling/heating slots



Stepper motor + spindle lowers assembly into dark vessel



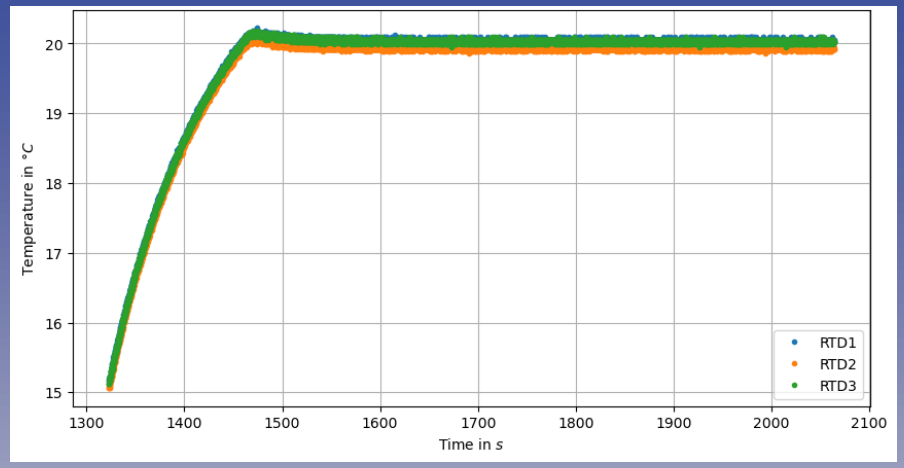
Quick release levers

SiPM temperature control electronics

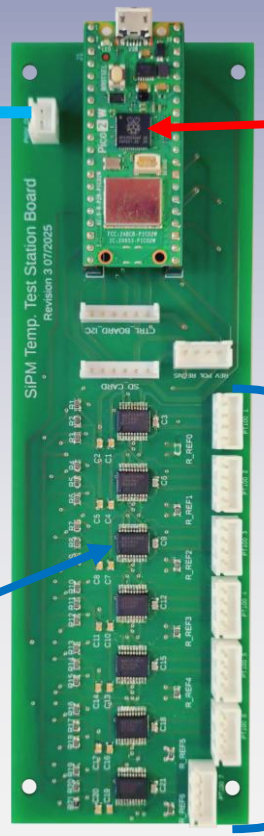


Custom Switching-PSU

DC output for Peltier-element



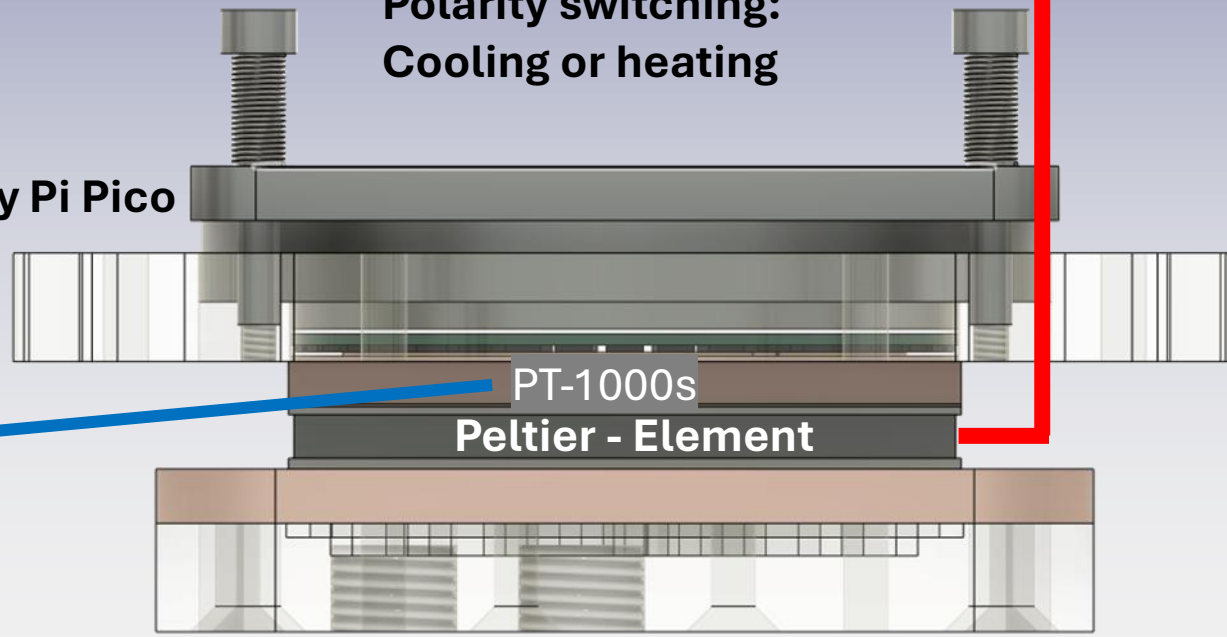
PWM



Mainboard:

PID feedback loop:
IN: Temp
OUT: PWM Duty Cycle
„The Brain:“ Raspberry Pi Pico
Low cost: 60€

Polarity switching:
Cooling or heating



Up to 7 PT-1000!
Read temperature

MAX31865
PT-1000
readout ICs

SiPM measurement and DAQ

SiPM Board

Amplifier board

CAEN SiPM power supply

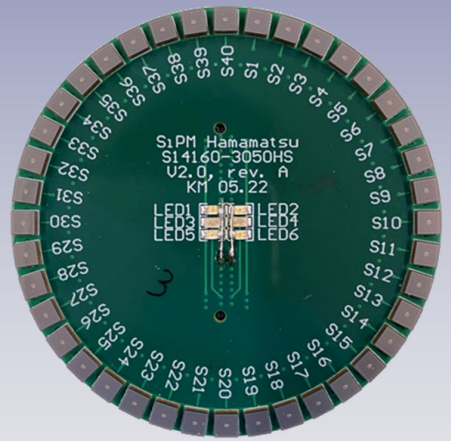
Mainboard

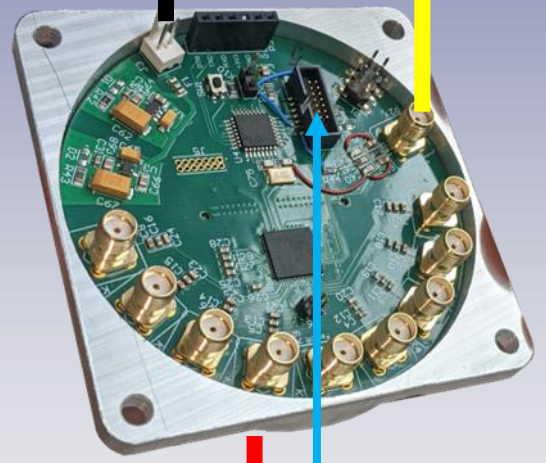
PC/RPi and remote access (WiFi)

Amplifier power

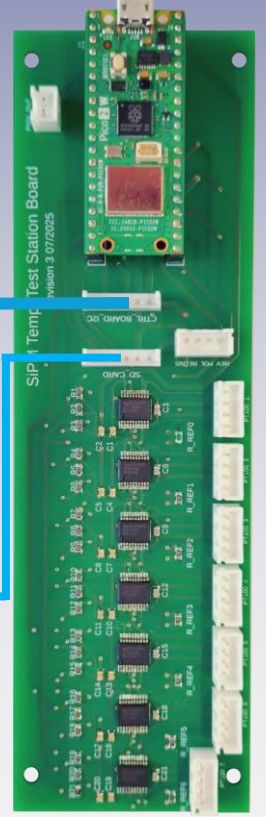
SiPM bias voltage

Transfer meas. data via UART











Saves data in CSV file

SiPMs connected to AMP in 8 Ch

Amplifier config

Sets bias voltage

Measures Current

Through all 40 SiPMs



Low Ripple 12V PSU

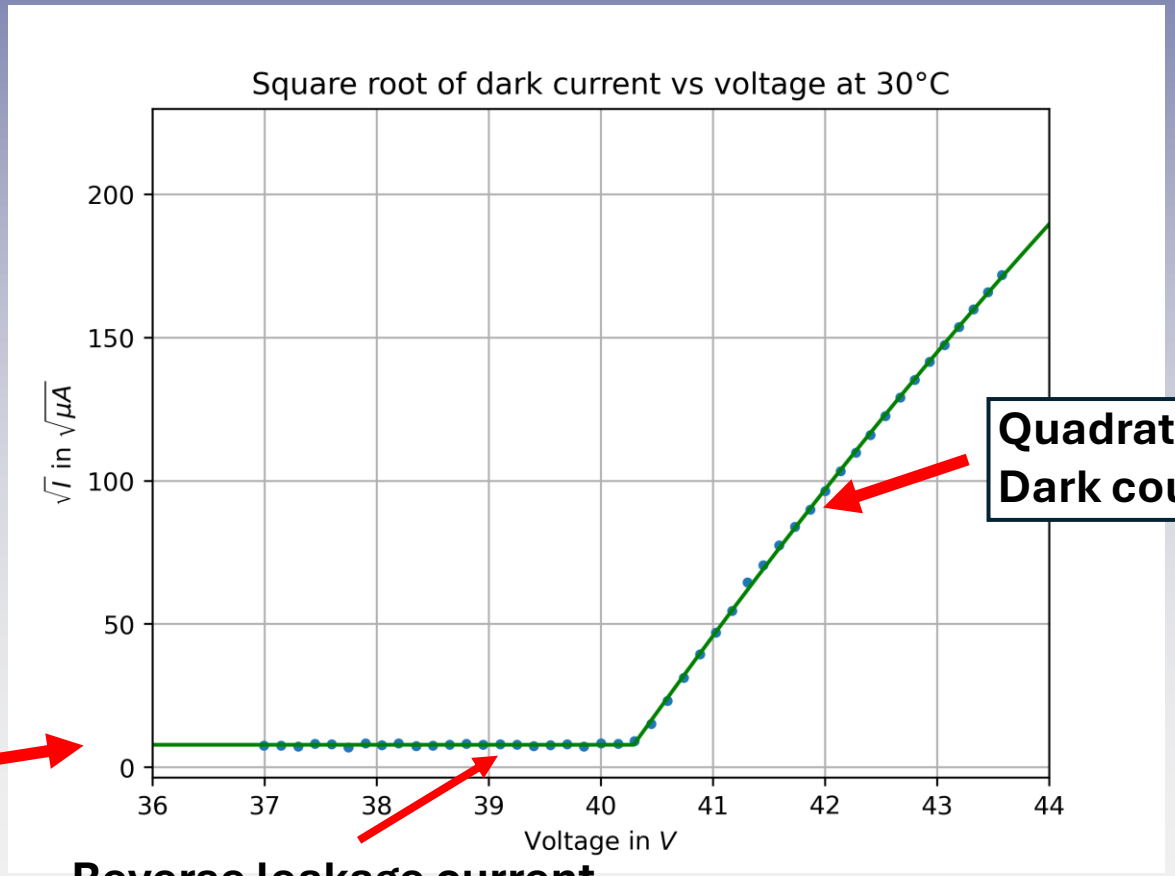
Finding breakdown voltages

1. Sweep bias voltages at fixed temperatures, measure SiPM current I
2. Fit piecewise functions to the √I-Voltage data
3. Find breakdown voltage as intersection with abscissa

$$f(V) = \begin{cases} c, & V \leq V_0 \\ a\,(V - V_0)^2 + b\,(V - V_0) + c, & V > V_0 \end{cases}$$

Quadratic fit

Expectation: linear rise of √I
Used: Quadratic -> account for residual resistances in circuit



Quadratic current rise:
Dark counts(∝V) x Gain(∝V)

For each temperature we measure this!

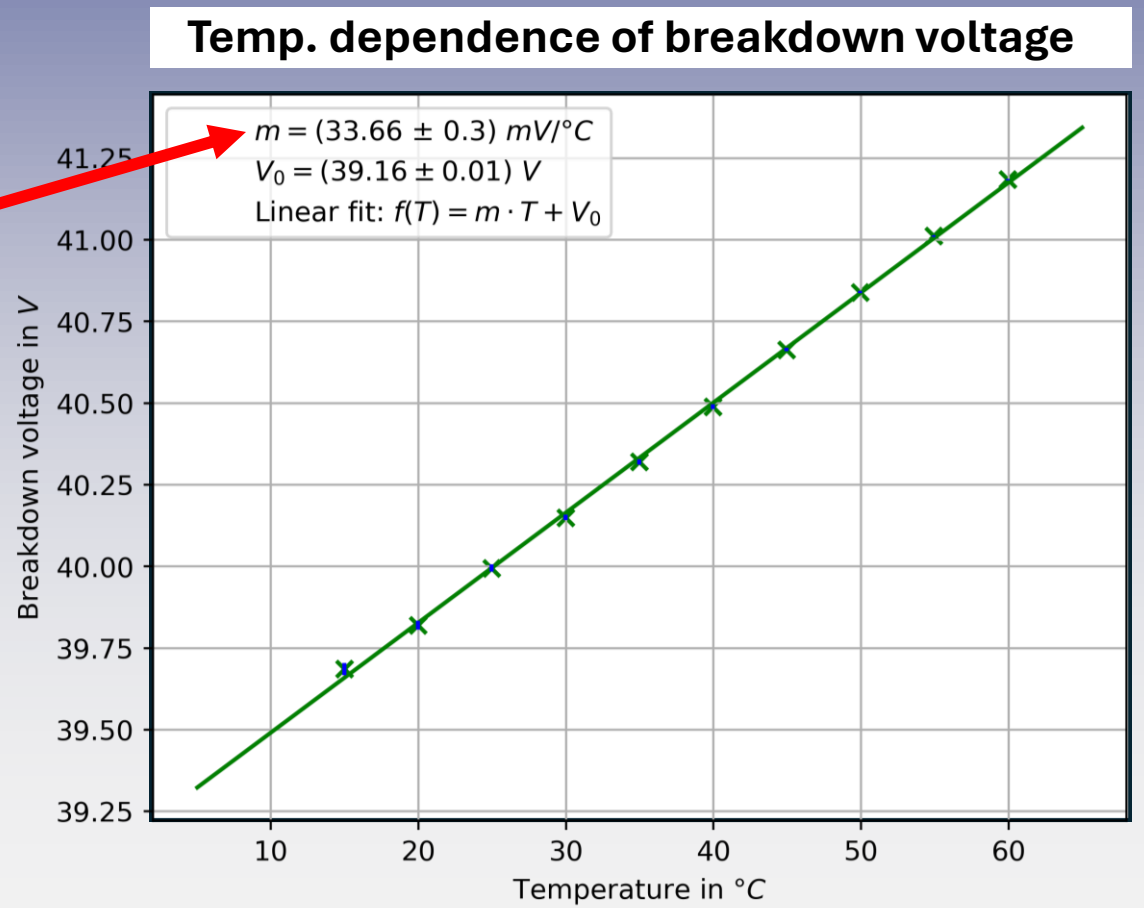
Reverse leakage current

Finding temperature dependence

1. Plot the intersections from the dark current fits against their temperature
2. Linear fit the intersection points
3. Resulting slope = temperature dependence of the breakdown voltage

We now know the change of V_b with temperature!
m = (33.66 ± 0.30) mV/K

In Experiment: Measure temperature and adjust bias voltage to stabilise gain!

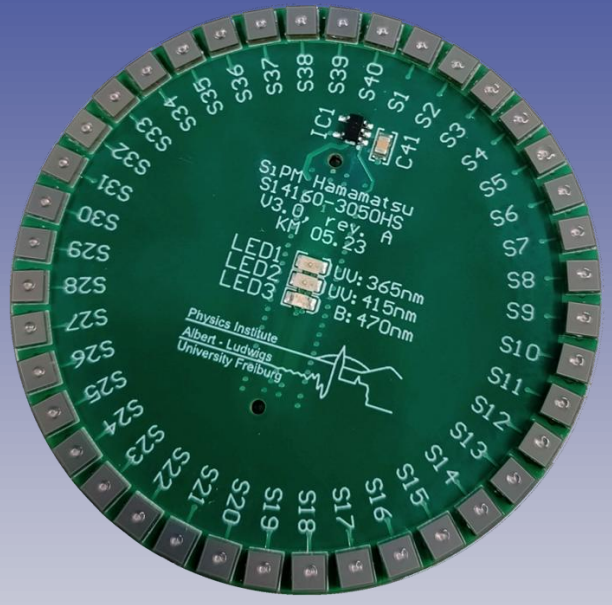


Each of our SiPM boards „gives“ one of these plots!

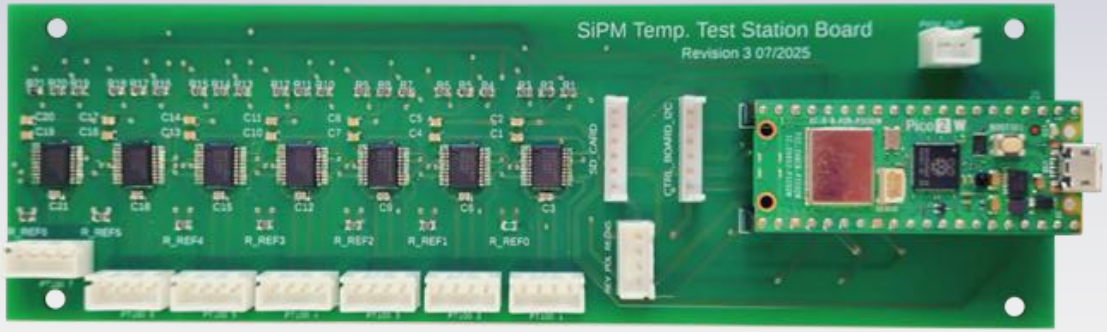
Thank you for your attention!

We found:

- **Epoxy has similar coupling performance to grease with better longterm stability!**
- **Temperature dependence of the SiPMs breakdown voltage to compensate for later!**
- **Developed cost effective electronics and mechanics to investigate SiPM behaviour in the dark!**



SHiP
Search for Hidden Particles



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